SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Classify the packings and explain in detail with a diagram.

Q.24 Calculate the time of drying for constant period & falling rate period.

Q.25 Describe cooling tower and their different types.

Roll No. ....

4th Sem / Chemical

Subject : Mass Transfer Operation - I

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 In mass transfer operation mass transfer may occur in

a) In one direction b) In opposite direction

c) Both a & b d) None of the above

Q.2 Molar flow rate means

a) Mass per unit time

b) Moles per unit time

c) Mass per unit area per unit time

d) Moles per unit area per unit time

Q.3 Packed column used for

a) Humidification b) Dehumidification

c) Absorption d) Desorption

Q.4 Saturation humidity is when air

a) Saturate b) Non saturate

c) Both a & b d) None of the above

Q.5 Cooling tower is used for cooling

a) Air b) liquid

c) Solids d) None of the above

Q.6 Drying refer to removal of

a) Liquid b) Air

c) Moisture d) Solid

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Diffusion due to random movement/ motion of the molecules is called

Q.8 Define mass transfer film coefficient.

Q.9 The other name of gas absorption is_______

Q.10 Define dew point.

Q.11 Define relative humidity.

Q.12 Define falling rate period.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain diffusivity and the factors which affect diffusivity.

Q.14 Describe Higbie's Penetration theory.

Q.15 Derive the equation for steady state diffusion through equimolecular counter diffusion.

Q.16 Write the properties for tower packing.

Q.17 Describe construction and working of mechanically agitated vessel with a diagram.

Q.18 Define humidity, humid heat, humid volume and saturate gas.

Q.19 Explain mechanism of humidifier with a diagram.

Q.20 Write advantages, disadvantages and application of tray Drier.

Q.21 Explain mechanism of drying.

Q.22 Give reasons for carried out drying operation.